



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

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Practical 7

AIM:- Perform DOM traversal and updates; develop a to - do list add, edit, and delete features.

Theory:-

DOM (Document Object Model) is a programming interface provided by the browser that represents a web page as a hierarchical tree of objects. It acts as a bridge, allowing JavaScript to access, modify, and delete a website's structure, content, and styling dynamically without needing to reload the page.

DOM Traversal (Moving Around the Tree)

DOM Traversal refers to navigating between elements based on their relationships in the family tree (Parents, Children, and Siblings)

DOM Updates (Modifying the Tree)

Once you have traversed to or selected the right element, DOM updating allows you to change its content, change its appearance, or restructure the page entirely

DOM Traversal Functions

Traversal means moving from one element to another in the HTML tree structure.

- **document.getElementById(id):** Finds a single element by its unique ID.
- **document.querySelector(selector):** Finds the first element that matches a CSS selector (like '.class' or '#id').
- **document.querySelectorAll(selector):** Finds all elements that match a CSS selector and returns a list.
- **element.parentElement:** Moves up to the parent container of an element.
- **element.children:** Gets a list of all direct child elements inside an element.

Adding Elements :-

Adding new content to a page requires creating the element and placing it inside the DOM.

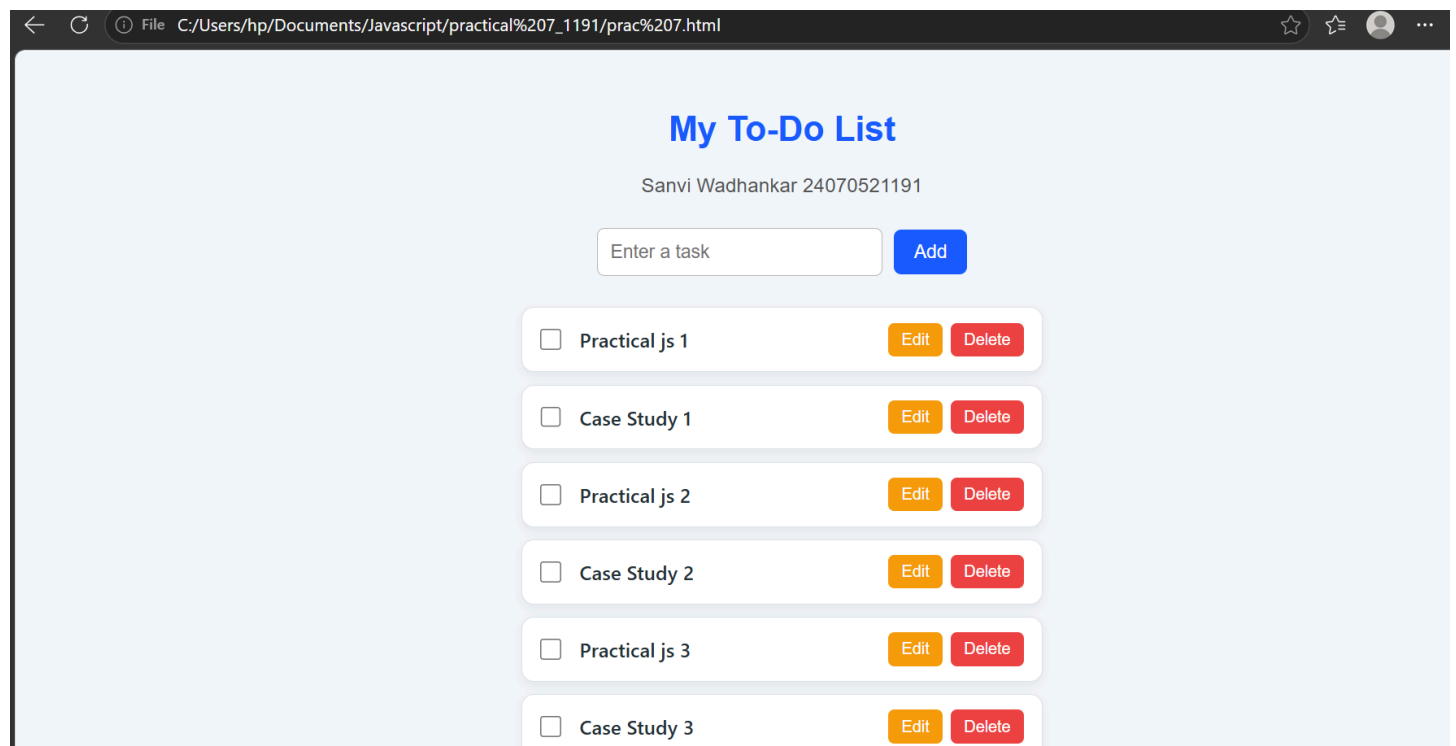
- **document.createElement(tagName)**: Creates a brand new HTML element in memory (like a div or li).
- **element.textContent = "text"**: Sets the safe text inside an element.
- **parent.appendChild(child)**: Adds the new child element to the end of a parent element.

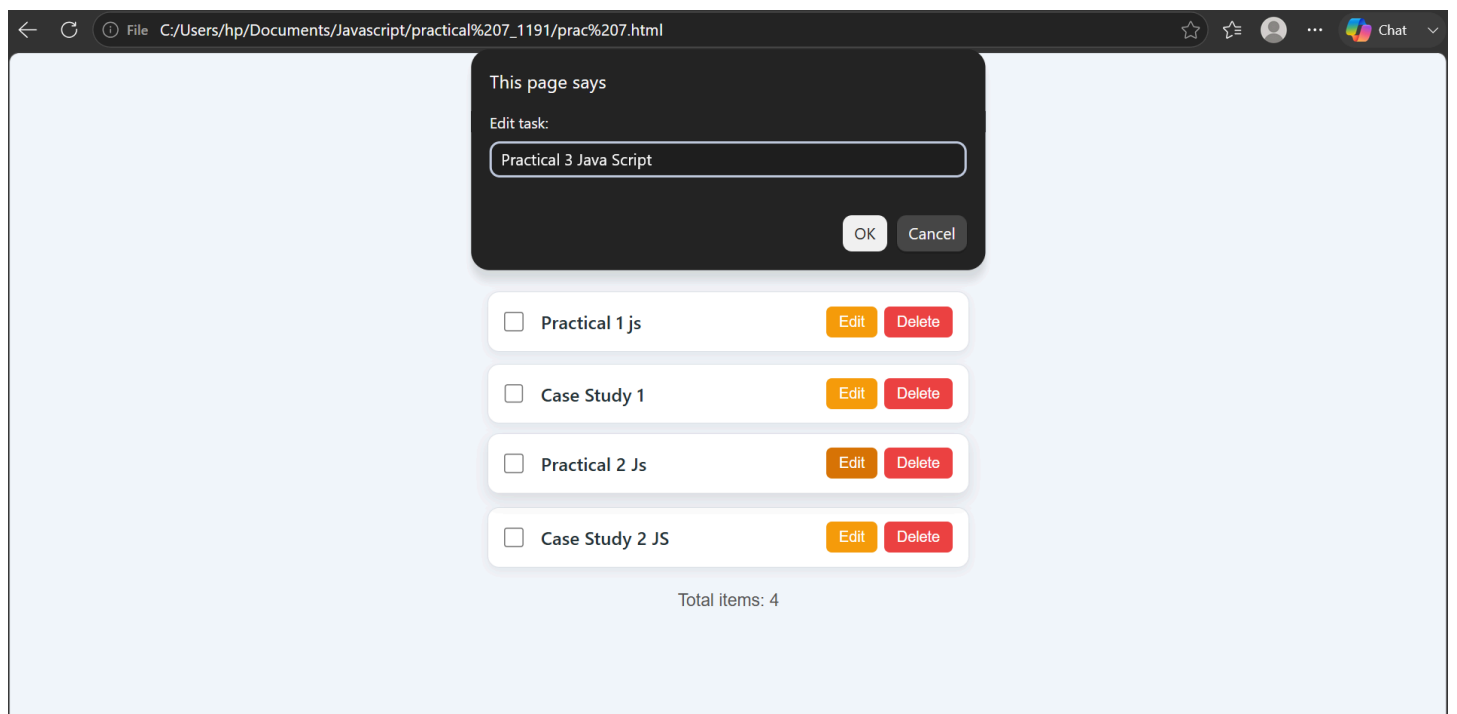
Editing Elements :-

Editing means changing the text, style, or attributes of an existing element.

- **element.textContent = "New Text"**: Updates the text inside an element.
- **element.innerHTML = "Bold"**: Replaces the HTML content inside an element.
- **element.style.color = "red"**: Changes a CSS style property directly.
- **element.classList.add('active')**: Adds a CSS class to an element.
- **element.classList.remove('active')**: Removes a CSS class from an element.

Output:-





Conclusion:-

Hence we have performed a to - do list using Document Object Model for updating , deleting , adding and editing list items in the to do list.